

Quiz 4 - Solutions

Exercise 1 (6 points)

Propose an objective function that incorporates class label information into PCA (i.e. supervised dimensionality reduction) such that samples of the same class are mapped to nearby locations in the new space. You do not have to solve for a solution, but must explain all terms and their function.

We can incorporate a term to the objective function to encourage samples from the same classes to be mapped near each other, and samples from different classes to be mapped far away from each other.

For example, consider,

$$J(\mathbf{U}) = \mathbf{U}^T \mathbf{R}_X \mathbf{U} - \lambda \sum_{i \in C_k} \sum_{j \in C_k} \|\mathbf{y}_i - \mathbf{y}_j\|_2^2$$

where $\mathbf{y}_i$ is the projection of sample $\mathbf{x}_i$ and λ is a hyperparameter. The optimization problem is thus,

$$\arg \max_{\mathbf{U}} J(\mathbf{U})$$

The second term computes the pairwise distance between all points in the same class. Maximizing $J(\mathbf{U})$ will *force* the second term to have a small value.

Exercise 2 (3 points)

Consider the data matrix $\mathbf{X}$ of size $3 \times N$, where N is the number of samples. The covariance matrix $\mathbf{K}$, of size 3×3 has 3 eigenvectors $v_1 = [-0.99, 0.09, 0]^T$, $v_2 = [0, 0, 1]^T$ and $v_3 = [-0.09, -0.99, 0]^T$ with eigenvalues $\lambda_1 = 0.98$, $\lambda_2 = 0.5$ and $\lambda_3 = 1.98$, respectively. Answer the following questions:

1. (1 point) **What linear transformation would you use to uncorrelate the data $\mathbf{X}$? Provide a numerical solution with the information provided. Justify your answer.**
2. (1 point) **Define the linear transformation (with all numerical entries) that Principal Component Analysis (PCA) would use to project this 3-dimensional**

data to a 2-dimensional representation.

3. (0.5 points) **What is the amount of explained variance of this 2-D projection? Justify.**
4. (0.5 points) **Let $\mathbf{Y}$ be the data (linear) transformation obtained by principal component transform of $\mathbf{X}$ onto a 2-dimensional space. What is resulting covariance matrix of transformed data $\mathbf{Y}$? Use a numerical answer and justify your answer.**

1. We can use PCA to uncorrelate the data by performing a change of basis linear transformation, using the eigenbasis of the covariance matrix.

The transformation can be written as

$$\mathbf{Y} = \mathbf{A}\mathbf{X}$$

where

$$\mathbf{A} = [\mathbf{v}_3 \quad \mathbf{v}_1 \quad \mathbf{v}_2]^T = \begin{bmatrix} -0.09 & -0.99 & 0 \\ -0.99 & 0.09 & 0 \\ 0 & 0 & 1 \end{bmatrix}^T$$

2. PCA will project the data $\mathbf{X}$ into a 2-D space where the two main directions correspond to the orthonormal eigenvectors of the covariance matrix of $\mathbf{X}$ with the largest eigenvalues, in particular, $\mathbf{v}_3$ and $\mathbf{v}_1$. The 2-D transformation can be written as

$$\mathbf{Y} = \mathbf{A}\mathbf{X}$$

where

$$\mathbf{A} = [\mathbf{v}_3 \quad \mathbf{v}_1]^T = \begin{bmatrix} -0.09 & -0.99 \\ -0.99 & 0.09 \\ 0 & 0 \end{bmatrix}^T$$

3. The amount of explained variance corresponds to the sum of the eigenvalues of the directions of projection:

$$\frac{1.98 + 0.98}{1.98 + 0.98 + 0.5} \approx 0.8555 \Rightarrow 85.55\%$$

4. The covariance of the PCA projection is a diagonal matrix (uncorrelated features), and the elements along the diagonal corresponds to the eigenvalues of the eigenvectors of projection:

$$\text{cov}(\mathbf{Y}) = \begin{bmatrix} 1.98 & 0 \\ 0 & 0.98 \end{bmatrix}$$

